Perfect Skewness of Free Stable Densities

An equal-level curvature comparison and a global midpoint theorem

Research manuscript prepared with ChatGPT for Keio

8 November 2026

Version 1.0: complete analytic proof draft for external review

Abstract

We prove that every nondegenerate free stable density is perfectly skew: the difference between its values at two points equidistant from its mode has a constant sign on the entire positive half-line. For the strictly stable normalization with parameters (α, ρ), the sign is exactly $\operatorname{sgn}((1-\alpha)(\rho-\frac{1}{2}))$. This gives an affirmative answer to the question in Hasebe–Simon–Wang, Remark 13 (2020), and also determines the direction of the comparison. We obtain quantitative inequalities for the midpoints of all density level sets. A separate logarithmic argument treats every non-strict free stable law of index one. The main estimate compares the curvature of an angular log-density potential at equal levels. Integration of squared slopes transfers this comparison to an exact differential identity for the physical midpoint. The proof is analytic throughout, including the extremal admissible parameters and finite support edges. The boundary coordinates and basic unimodality ingredients are credited to prior work. Historical priority is not inferred merely from the absence of a solution in our literature search.

Keywords. Free stable law; perfect skewness; level set; mode; curvature comparison.

1 The original question and the results

For a continuous unimodal density f with unique mode M, perfect skewness means that

$$f(M+t) - f(M-t), \qquad t > 0, \tag{1.1}$$

has a constant sign, with the identically zero case allowed. The density is extended by zero outside its support. We call the skewness strict when the comparison is strict for every $t > 0$.

Hasebe, Simon and Wang ask whether this property holds for the full free strictly stable family in [3, Remark 13, p. 321]. The question already appears as Remark 18 in the May 2018 preprint [4]. Their published remark records several special cases and leaves the general case open. We retain exactly the density comparison (1.1). In particular, the midpoint quantities introduced below are proof devices, not replacements for the original target.

Write

$$G_\mu(z) = \int_{\mathbb{R}} \frac{\mu(\mathrm{d}x)}{z-x}, \qquad F_\mu(z) = \frac{1}{G_\mu(z)}, \qquad \phi_\mu(z) = F_\mu^{-1}(z) - z.$$

The inverse is initially understood near infinity in the upper half-plane. We use the standard free strictly stable normalization

$$\phi_{\alpha,\rho}(z) = -e^{i\pi\alpha\rho} z^{1-\alpha}, \qquad z \in \mathbb{C}^+, \tag{1.2}$$

with the principal branch of the power and admissible parameters

$$\mathcal{A} = \{0 < \alpha \le 1,\ 0 \le \rho \le 1\} \ \cup\ \{1 < \alpha \le 2,\ 1 - 1/\alpha \le \rho \le 1/\alpha\}. \tag{1.3}$$

See [1, 2, 3] for the classification and conventions. The pairs $(1,0)$ and $(1,1)$ give point masses and are excluded from all density statements.

Theorem 1.1 (Global comparison in the original normalization)**.** *For every nondegenerate admissible (α,ρ), let $f_{\alpha,\rho}$ be the density associated with (1.2), and let $M_{\alpha,\rho}$ be its unique mode. Then, for every $t>0$,*

$$\boxed{\operatorname{sgn}(f_{\alpha,\rho}(M_{\alpha,\rho}+t)-f_{\alpha,\rho}(M_{\alpha,\rho}-t))=\operatorname{sgn}\Big((1-\alpha)(\rho-\tfrac{1}{2})\Big).} \tag{1.4}$$

In particular, the comparison is strict whenever $\alpha\neq 1$ and $\rho\neq\frac{1}{2}$. If $\rho=\frac{1}{2}$, or if $\alpha=1$ and the law is nondegenerate, the density is symmetric about its mode.

For each density considered here, every $0<h<h_*:=f(M)$ has exactly two points of height h:

$$x_L(h)<M<x_R(h), \qquad f(x_L(h))=f(x_R(h))=h.$$

Let

$$C(h)=\frac{x_L(h)+x_R(h)}{2}. \tag{1.5}$$

Theorem 1.2 (Quantitative midpoint comparison)**.** *In the normalization (1.2), suppose $0<\alpha<2$, $\alpha\neq 1$, and $\rho>\frac{1}{2}$ is admissible. Then $M:=M_{\alpha,\rho}>0$, and, for every $0<h<h_*$,*

$$1<\alpha<2: \quad C(h)<M(h/h_*)^{\alpha-1}<M, \tag{1.6}$$

$$0<\alpha<1: \quad C(h)>M(h/h_*)^{\alpha-1}>M. \tag{1.7}$$

More precisely, $h^{1-\alpha}C(h)$ is strictly increasing if $\alpha>1$ and strictly decreasing if $\alpha<1$. Reflection gives the corresponding statements for $\rho<\frac{1}{2}$.

The next result includes the free stable laws of index one which are not strictly stable. The classification in [1, Appendix], also described in [3, p. 298], reduces these laws, after reflection and a real translation, to the logarithmic form below.

Theorem 1.3 (The complete logarithmic index-one family)**.** *Let μ have Voiculescu transform*

$$\phi_\mu(z)=c-iA-B\log z, \qquad A\geq 0, \quad B>0, \quad c\in\mathbb{R}, \quad z\in\mathbb{C}^+, \tag{1.8}$$

where log *is principal. Its density has a unique mode M and satisfies*

$$f(M+t)<f(M-t) \qquad \text{for every } t>0. \tag{1.9}$$

Moreover its level-set midpoint satisfies

$$C(h)<M+B\log(h/h_*)<M, \qquad 0<h<h_*. \tag{1.10}$$

The reflected laws have the opposite strict comparison. When $B=0$ and $A>0$, the density is a translated Cauchy density and the comparison is equality.

Corollary 1.4 (All free stable laws)**.** *Every nondegenerate free stable density is perfectly skew about its unique mode. The comparison is strict unless the density is symmetric about the mode.*

Reduction of Corollary 1.4 to the theorems. The classification [1, Appendix] gives, up to affine changes, (1.2) for indices other than one and the Cauchy or logarithmic cases for index one. If $g(x)=\lambda^{-1}f((x-d)/\lambda)$ with $\lambda>0$, its mode is $m=d+\lambda M$ and

$$g(m+t)-g(m-t)=\lambda^{-1}\{f(M+t/\lambda)-f(M-t/\lambda)\}.$$

Thus positive affine changes preserve the comparison, while reflection reverses its direction. Theorems 1.1 and 1.3 therefore give the assertion. □

2 An auxiliary equal-level comparison problem

We first isolate the mechanism that will apply to both a sine function and an affine function. The auxiliary problem is to compare the curvatures of a strictly convex potential at its two points of equal height, and then to compare the magnitudes of its two slopes.

Lemma 2.1 (Equal-level curvature comparison)**.** *Let $\beta > 1$, $Q \geq 0$, and let $L : (0, \pi) \to (0, \infty)$ be C^2. Suppose*

$$(L')^2 - LL'' = Q, \qquad L(\theta) \geq \sqrt{Q}\,\sin\theta \quad (0 < \theta < \pi). \tag{2.1}$$

Set

$$V(\theta) = \log L(\theta) - \beta \log\sin\theta. \tag{2.2}$$

Then $V'' \geq (\beta - 1)\csc^2\theta > 0$, V tends to $+\infty$ at both ends, and it has a unique minimum at some θ_m. If $\theta_1 < \theta_2$ satisfy $V(\theta_1) = V(\theta_2)$ and $r = \sin\theta_2/\sin\theta_1$, then

$$\operatorname{sgn}(V''(\theta_1) - V''(\theta_2)) = \operatorname{sgn}(\log r). \tag{2.3}$$

For $r > 1$ the following quantitative lower bound holds, with $z = r^{-2}$:

$$V''(\theta_1) - V''(\theta_2) \geq \csc^2\theta_1\big[\beta(1 - z) - (1 - z^\beta)\big] > 0. \tag{2.4}$$

Proof. The first identity in (2.1) gives $(\log L)'' = -Q/L^2$. Hence

$$V'' = \beta\csc^2\theta - \frac{Q}{L(\theta)^2} \geq (\beta - 1)\csc^2\theta > 0. \tag{2.5}$$

If $Q > 0$, the lower bound for L gives $V \geq \log\sqrt{Q} + (1 - \beta)\log\sin\theta$, which tends to $+\infty$ at each endpoint. If $Q = 0$, then $\log L$ is affine in θ, so the same endpoint conclusion follows directly. Strict convexity now gives the unique minimum.

At equal levels,

$$L(\theta_2) = r^\beta L(\theta_1). \tag{2.6}$$

Suppose first that $r > 1$, and put $z = r^{-2} \in (0, 1)$. Then

$$\begin{aligned} V''(\theta_1) - V''(\theta_2) &= \beta\csc^2\theta_1(1 - z) - \frac{Q}{L(\theta_1)^2}(1 - z^\beta) \\ &\geq \csc^2\theta_1\{\beta(1 - z) - (1 - z^\beta)\}. \end{aligned}$$

The bracket is strictly positive because the function $z \mapsto z^\beta$ is strictly convex: $z^\beta > 1 + \beta(z - 1)$ for $z \neq 1$. If $r < 1$, exchange indices and apply this argument with $1/r > 1$, using the bound for L at θ_2. This gives the reverse strict comparison. If $r = 1$, (2.6) gives equality of the two curvatures. □

Lemma 2.2 (From curvature to slope imbalance)**.** *Under the assumptions of Lemma 2.1, set $u_m = V(\theta_m)$ and $S = -V'$. For $u > u_m$, denote the inverse branches of V by $\theta_1(u) < \theta_m < \theta_2(u)$, and write $S_i(u) = S(\theta_i(u))$. Suppose the sign*

$$\operatorname{sgn}\log\frac{\sin\theta_2(u)}{\sin\theta_1(u)} = \varepsilon \in \{-1, 1\}$$

is the same for every $u > u_m$. Then

$$S_1 > 0 > S_2, \qquad \operatorname{sgn}(S_1 + S_2) = \varepsilon \quad \textit{at every level.} \tag{2.7}$$

Proof. The signs of the slopes follow from strict convexity. Differentiating along $V(\theta_i(u)) = u$ gives

$$\frac{\mathrm{d}}{\mathrm{d}u} S_i(u)^2 = 2V''(\theta_i(u)). \tag{2.8}$$

Both squared slopes tend to zero as $u \downarrow u_m$. Therefore

$$S_1(u)^2 - S_2(u)^2 = 2\int_{u_m}^{u} \{V''(\theta_1(w)) - V''(\theta_2(w))\}\,\mathrm{d}w. \tag{2.9}$$

The integrand is continuous at the lower limit after taking its limit there, and has strict sign ε at every larger level by Lemma 2.1. The integral has that sign as well. Finally $S_1 - S_2 > 0$, so $S_1^2 - S_2^2 = (S_1 - S_2)(S_1 + S_2)$ proves the claim. □

Lemma 2.3 (Level midpoints imply the pointwise comparison)**.** *Let f be continuous, positive on the interior of an interval, strictly increasing up to its mode M and strictly decreasing after it, and tending to zero at both ends of that interval. Define $C(h)$ by (1.5). If the left support endpoint is $-\infty$ and $C(h) < M$ at every $0 < h < f(M)$, then $f(M + t) < f(M - t)$ for every $t > 0$. If the right support endpoint is $+\infty$ and $C(h) > M$ at every such level, the reverse strict comparison holds.*

Proof. Consider the first assertion. If $M + t$ lies in the interior of the support, let $h = f(M + t)$. Then $x_R(h) = M + t$ and $x_L(h) < M - t < M$. Strict increase on the left branch gives $f(M - t) > h$. If $M + t$ is at or beyond a finite right endpoint, its density is zero while $f(M - t) > 0$, since the left endpoint is $-\infty$. This includes every $t > 0$. Reflection proves the second assertion. □

3 Boundary coordinates and the angular potential

We now work with $0 < \alpha < 2$, $\alpha \neq 1$. Put

$$p = \alpha - 1, \qquad \beta = \alpha + 1 = p + 2, \qquad A = \pi\alpha\rho, \qquad q = A - \frac{\pi\alpha}{2}. \tag{3.1}$$

Reflection replaces ρ by $1 - \rho$; see [2, 3]. It suffices for the asymmetric case to assume $\rho > \frac{1}{2}$, so $0 < q < \pi/2$.

3.1 The standard boundary parametrization

For $0 < \theta < \pi$, define

$$v = A - p\theta, \qquad R = \left(\frac{\sin v}{\sin\theta}\right)^{1/\alpha}, \qquad x(\theta) = R(\cos\theta - \sin\theta\cot v), \qquad y(\theta) = \frac{\sin\theta}{\pi R}. \tag{3.2}$$

Admissibility gives $0 < v < \pi$. These are the free stable boundary coordinates of Biane [1, Appendix A.1], in the convention of [2]. In particular, $x(\theta)$ traverses the support interior and $f_{\alpha,\rho}(x(\theta)) = y(\theta)$. To check the formulas in our normalization, set

$$K(w) = w - e^{iA}w^{-p}, \qquad w = Re^{i\theta}.$$

The equation $\operatorname{Im} K(w) = 0$ gives $R^\alpha = \sin v/\sin\theta$, its real part gives x, and Stieltjes inversion gives $y = -\operatorname{Im}(1/w)/\pi$. The identification with the density, including the support endpoints, is the standard boundary parametrization just cited.

Differentiation gives

$$\frac{R'}{R} = -\frac{\cot\theta + p\cot v}{\alpha}, \qquad x'(\theta) = -\frac{R\sin\theta}{\alpha}\left\{(\cot\theta + p\cot v)^2 + \alpha^2\right\} < 0. \tag{3.3}$$

We will use this strict decrease to distinguish the physical left branch from the angular left branch.

3.2 A sine bound and strict convexity

The elementary sine estimate

$$\sin v \geq |p| \sin\theta \tag{3.4}$$

holds on the whole admissible parameter range under consideration. Indeed, if $p > 0$, then

$$p(\pi - \theta) \leq v \leq \pi - p\theta.$$

If $p < 0$, write $s = -p \in (0, 1)$; then

$$s\theta \leq v \leq \pi - s(\pi - \theta).$$

In either interval the minimum of sine is at an endpoint, and $\sin(su) \geq s \sin u$ for $0 \leq s \leq 1$, $0 \leq u \leq \pi$, by concavity of sine. This proves (3.4).

Set

$$V(\theta) = \log\sin v - \beta \log\sin\theta = -\alpha \log(\pi y(\theta)), \qquad S(\theta) = -V'(\theta) = \beta\cot\theta + p\cot v. \tag{3.5}$$

The function $L(\theta) = \sin(A - p\theta)$ satisfies $(L')^2 - LL'' = p^2$. Thus Lemma 2.1 applies with $Q = p^2$, and yields

$$V'' = \beta\csc^2\theta - p^2\csc^2 v \geq \alpha\csc^2\theta > 0. \tag{3.6}$$

Consequently y tends to zero at both angular endpoints, has a unique maximum at θ_m, and is strictly increasing before θ_m and strictly decreasing after it. In combination with $x' < 0$, this proves strict unimodality in the physical coordinate.

The coordinates, the sine bound for $\alpha > 1$, and the resulting unimodality estimate also occur in corpus manuscript [122] [7, Section 2]. They are not claimed as new here. The additional ingredient is the comparison at two different equal-level points in (2.4), followed by (2.9) and the midpoint identity below.

3.3 The mode and the support side needed globally

At the central angle,

$$V'(\pi/2) = p\tan q.$$

Since V' is strictly increasing, $\theta_m < \pi/2$ if $p > 0$ and $\theta_m > \pi/2$ if $p < 0$. At the mode, $S(\theta_m) = 0$, so

$$M = x(\theta_m) = \frac{2\alpha}{p} R(\theta_m)\cos\theta_m > 0. \tag{3.7}$$

For later use, if $p > 0$ and $\rho > \frac{1}{2}$, then $A > \pi\alpha/2 > p\pi$. As $\theta \uparrow \pi$, $\sin v$ has a positive limit, $R \to \infty$, and $x(\theta) \to -\infty$. Thus the left support endpoint is $-\infty$. The right endpoint is $+\infty$ unless $A = \pi$, when it is $\alpha p^{1/\alpha - 1}$. If $p < 0$ and $\rho > \frac{1}{2}$, then $A > 0$ and $A \leq \pi\alpha < \pi$, so as $\theta \downarrow 0$ we have $R \to \infty$ and $x(\theta) \to +\infty$. Thus the right support endpoint is $+\infty$. The left endpoint is finite only when $\rho = 1$, in which case it is $\alpha(1-\alpha)^{1/\alpha - 1}$. At each finite endpoint $y(\theta) \to 0$. These facts verify the support hypotheses of Lemma 2.3, including the extremal parameters.

4 Proof of the global strictly stable comparison

4.1 Orienting the equal-level angles

Fix $0 < h < h_*$. There are exactly two angles

$$\theta_1 < \theta_m < \theta_2, \qquad y(\theta_1) = y(\theta_2) = h.$$

Because $x' < 0$, they represent $x_R(h) = x(\theta_1)$ and $x_L(h) = x(\theta_2)$, respectively. We first prove

$$\operatorname{sgn} \log \frac{\sin \theta_2}{\sin \theta_1} = \operatorname{sgn} p. \tag{4.1}$$

For $0 < t < \pi/2$, the formulas above give

$$\big(\pi y(\pi/2 \pm t)\big)^{\alpha} = \frac{\cos^{\beta} t}{\cos(q \mp pt)}. \tag{4.2}$$

Both denominators are positive by admissibility. Their difference is

$$\cos(q - pt) - \cos(q + pt) = 2 \sin q \sin(pt),$$

which has the sign of p.

If $p > 0$, it follows that $y(\pi/2 + t) < y(\pi/2 - t)$. Here $\theta_1 < \theta_m < \pi/2$. If $\theta_1 + \theta_2 \geq \pi$, then $\theta_2 \geq \pi - \theta_1 > \theta_m$, and strict decrease to the right of the angular mode would give

$$y(\theta_2) \leq y(\pi - \theta_1) < y(\theta_1),$$

a contradiction. Hence $\theta_1 + \theta_2 < \pi$. If $p < 0$, the inequality in (4.2) reverses and $\theta_2 > \theta_m > \pi/2$. Were $\theta_1 + \theta_2 \leq \pi$, then $\theta_1 \leq \pi - \theta_2 < \theta_m$ and strict increase on the left would give $y(\theta_1) \leq y(\pi - \theta_2) < y(\theta_2)$, again a contradiction. Thus $\theta_1 + \theta_2 > \pi$ in this case. Finally,

$$\sin \theta_2 - \sin \theta_1 = 2 \cos \frac{\theta_1 + \theta_2}{2} \sin \frac{\theta_2 - \theta_1}{2}$$

proves (4.1).

4.2 A strict curvature gap at every level

Equality of the density heights gives the exact relation

$$\sin v_2 = r^{\beta} \sin v_1, \qquad r = \frac{\sin \theta_2}{\sin \theta_1}. \tag{4.3}$$

For $r > 1$, Lemma 2.1 specializes to

$$\begin{aligned} V''(\theta_1) - V''(\theta_2) &\geq \csc^2 \theta_1 \left[\beta(1 - r^{-2}) - (1 - r^{-2\beta})\right] \\ &> 0. \end{aligned} \tag{4.4}$$

For $r < 1$, exchange the indices as in that lemma. Together with (4.1), this proves

$$\operatorname{sgn}(V''(\theta_1) - V''(\theta_2)) = \operatorname{sgn} p$$

at every nonmaximal density level. Lemma 2.2 now gives

$$S_1 > 0 > S_2, \qquad \operatorname{sgn}(S_1 + S_2) = \operatorname{sgn} p, \qquad S_i = S(\theta_i). \tag{4.5}$$

This is the strict global estimate needed to control the physical midpoints. It does not require an expansion near the mode or in a tail.

4.3 The exact midpoint differential identity

From (3.5), $y' = yS/\alpha$. Combining this identity with (3.3), along either branch of height h we get

$$\frac{\mathrm{d}x}{\mathrm{d}h} = -\frac{D(\theta)}{\pi h^2}, \qquad D(\theta) = \frac{\sin^2\theta\{(\cot\theta + p\cot v)^2 + \alpha^2\}}{S(\theta)}. \tag{4.6}$$

Since $\cot\theta + p\cot v = S - \alpha\cot\theta$, division gives

$$\begin{aligned} D(\theta) &= \sin^2\theta\{S - 2\alpha\cot\theta\} + \frac{\alpha^2}{S} \\ &= p\sin^2\theta(\cot v - \cot\theta) + \frac{\alpha^2}{S} = -p\pi h\, x(\theta) + \frac{\alpha^2}{S}. \end{aligned} \tag{4.7}$$

Here the last equality uses $\pi h\, x(\theta) = \sin^2\theta(\cot\theta - \cot v)$. Averaging (4.6) over the two branches yields the exact identity

$$\boxed{C'(h) - \frac{p}{h}C(h) = -\frac{\alpha^2}{2\pi h^2}\left(\frac{1}{S_1} + \frac{1}{S_2}\right).} \tag{4.8}$$

Because $S_1 S_2 < 0$, equation (4.5) shows that the right-hand side has strict sign p. Multiplication by the positive integrating factor h^{-p} gives

$$\operatorname{sgn}\frac{\mathrm{d}}{\mathrm{d}h}\{h^{-p}C(h)\} = \operatorname{sgn} p. \tag{4.9}$$

As $h \uparrow h_*$, both inverse branches converge to the mode and $C(h) \to M$. Strict monotonicity in (4.9) and this finite endpoint limit imply

$$\begin{cases} C(h) < M(h/h_*)^p, & p > 0, \\ C(h) > M(h/h_*)^p, & p < 0. \end{cases}$$

The inequalities are strict: between any fixed $h < h_*$ and the endpoint there is another interior level at which strict monotonicity already gives a positive difference. Combining these inequalities with $M > 0$ from (3.7) proves Theorem 1.2. No integration across a possibly singular branch derivative at $h = h_*$ is needed.

4.4 Every distance, all parameters, and the degenerate corners

For $\rho > \frac{1}{2}$ and $\alpha > 1$, (1.6) gives $C(h) < M$ at every level, and the left support endpoint is $-\infty$. Lemma 2.3 gives $f(M + t) < f(M - t)$ for every $t > 0$, including points beyond a finite right endpoint. For $\rho > \frac{1}{2}$ and $\alpha < 1$, (1.7) and the infinite right endpoint give $f(M + t) > f(M - t)$ for every $t > 0$. Reflection proves the cases $\rho < \frac{1}{2}$.

When $\rho = \frac{1}{2}$, the law is symmetric. At $\alpha = 1$ and $0 < \rho < 1$, the transform (1.2) gives

$$f_{1,\rho}(x) = \frac{\sin(\pi\rho)}{\pi\{(x + \cos(\pi\rho))^2 + \sin^2(\pi\rho)\}},$$

which is symmetric about its mode $-\cos(\pi\rho)$. The two remaining index-one corners are point masses, already excluded. At $\alpha = 2$ the only admissible value is $\rho = \frac{1}{2}$, and the law is the symmetric semicircle. This proves Theorem 1.1 on its entire stated domain. □

5 The logarithmic index-one family

5.1 Coordinates and support

For (1.8), put

$$\begin{aligned} T(\theta) &= A + B\theta, \qquad R(\theta) = \frac{T(\theta)}{\sin\theta}, \\ x(\theta) &= T(\theta)\cot\theta - B\log R(\theta) + c, \qquad y(\theta) = \frac{\sin^2\theta}{\pi T(\theta)}, \qquad 0 < \theta < \pi. \end{aligned} \tag{5.1}$$

We use the boundary method of [1, Appendix, Proposition A1.3], and fix the logarithmic sign directly from the transform in (1.8). Indeed, with $w = Re^{i\theta}$ and $K(w) = w + \phi_\mu(w)$,

$$\operatorname{Im} K(w) = R\sin\theta - A - B\theta, \qquad \operatorname{Re} K(w) = R\cos\theta - B\log R + c.$$

Setting the imaginary part to zero, taking the real part, and applying Stieltjes inversion gives exactly (5.1). Direct differentiation then gives

$$x'(\theta) = -T(\theta)\left\{\left(\cot\theta - \frac{B}{T(\theta)}\right)^2 + 1\right\} < 0. \tag{5.2}$$

The density identification can also be justified directly. Define

$$\Omega = \{Re^{i\theta} : 0 < \theta < \pi,\ R > (A + B\theta)/\sin\theta\}.$$

This is connected, $K(\Omega) \subset \mathbb{C}^+$, and $|w| > B$ throughout Ω. Every finite boundary point maps to the real line, and $K(w) = w + O(\log|w|)$ at infinity. Hence $K : \Omega \to \mathbb{C}^+$ is proper: a sequence leaving all compact subsets can neither approach a finite boundary point nor tend to infinity while its image stays in a compact subset of $\mathbb{C}^+$. Its image is open and closed in $\mathbb{C}^+$, hence is all of $\mathbb{C}^+$. Since $K'(w) = 1 - B/w$ has no zero in Ω, this proper map is an unbranched covering of the simply connected upper half-plane, and is therefore a conformal bijection. Its inverse agrees near infinity with F_μ and hence agrees throughout $\mathbb{C}^+$. The boundary formulas and Stieltjes inversion thus identify the density without importing a sign convention for the logarithmic term from another displayed parametrization.

The left endpoint is always $-\infty$. The right endpoint is $+\infty$ if $A > 0$, and

$$B - B\log B + c \tag{5.3}$$

if $A = 0$. In the latter case the remaining real boundary ray $w \geq B$ maps to $[B - B\log B + c, \infty)$ under $w \mapsto w - B\log w + c$ and has zero density. The curved boundary in (5.1) covers the support interior. In both cases y tends to zero at each end.

5.2 The same curvature comparison

Here take $L(\theta) = T(\theta)$, $Q = B^2$, and $\beta = 2$ in Lemma 2.1. Its hypotheses hold because

$$(T')^2 - TT'' = B^2, \qquad T(\theta) = A + B\theta \geq B\theta \geq B\sin\theta.$$

Thus

$$V = \log T - 2\log\sin\theta = -\log(\pi y), \qquad S = -V' = 2\cot\theta - \frac{B}{T}, \qquad V'' = 2\csc^2\theta - \frac{B^2}{T^2} > 0. \tag{5.4}$$

There is a unique angular maximum of y, and (5.2) gives a unique physical mode M.

At equal density height, with $\theta_1 < \theta_m < \theta_2$,

$$\frac{T(\theta_2)}{T(\theta_1)} = \left(\frac{\sin\theta_2}{\sin\theta_1}\right)^2 = r^2.$$

Since $B > 0$ and $\theta_2 > \theta_1$, the left-hand side exceeds one, so $r > 1$ at every level. The curvature gap has the particularly simple form

$$V''(\theta_1) - V''(\theta_2) \geq \csc^2\theta_1(1 - r^{-2})^2 > 0. \tag{5.5}$$

Lemma 2.2 gives $S_1 > 0 > S_2$ and $S_1 + S_2 > 0$.

5.3 A logarithmic midpoint bound and reflection

Since $y' = yS$, equations (5.1) and (5.2) give

$$\frac{\mathrm{d}x}{\mathrm{d}h} = -\frac{D(\theta)}{\pi h^2}, \qquad D(\theta) = \frac{\sin^2\theta\{(\cot\theta - B/T)^2 + 1\}}{S}.$$

Now $\cot\theta - B/T = S - \cot\theta$, so division gives

$$D(\theta) = \sin^2\theta(S - 2\cot\theta) + \frac{1}{S} = -B\pi h + \frac{1}{S}. \tag{5.6}$$

Consequently

$$C'(h) = \frac{B}{h} - \frac{1}{2\pi h^2}\left(\frac{1}{S_1} + \frac{1}{S_2}\right) > \frac{B}{h}. \tag{5.7}$$

It follows that $C(h) - B\log h$ is strictly increasing. Taking its finite limit $M - B\log h_*$ at the top level gives $C(h) < M + B\log(h/h_*) < M$. The infinite left endpoint and Lemma 2.3 now give the strict comparison for every $t > 0$, including the finite right edge when $A = 0$.

For clarity, reflection must respect the logarithm branch. If $\phi_X(z) = c - iA - B\log z$ with $A \geq 0$, $B > 0$, then

$$\phi_{-X}(z) = -\overline{\phi_X(-\overline{z})} = -c - i(A + B\pi) + B\log z. \tag{5.8}$$

This follows from $\log(-\overline{z}) = \overline{\log z} + i\pi$ for $z \in \mathbb{C}^+$. Every admissible logarithmic law with the opposite coefficient is a reflection of this form. For $B = 0$, $A > 0$, the inverse transform gives a translated Cauchy density. This completes the proof of Theorem 1.3, and hence of Corollary 1.4. □

6 What the argument adds, and the limits of the claim

6.1 The substantive estimate

The method uses three steps: equal-level curvature comparison, integration of squared slopes, and an integrating factor for the physical midpoint. The strict remainder

$$\beta(1 - z) - (1 - z^\beta) > 0, \qquad 0 < z < 1, \quad \beta > 1,$$

is the sign-producing estimate. The angular change of variables by itself does not yield this comparison. Nor does strict unimodality alone determine which side of the mode has the larger density at every distance. Theorems 1.2 and 1.3 also quantify the displacement of all equal-height interval midpoints; no optimality of these quantitative bounds is asserted.

For index one, the usual representation $a\mathcal{C} + b\mathcal{T}$, with $a \geq 0$, $b \in \mathbb{R}$, where $\mathcal{C}$ is standard Cauchy, $\phi_{\mathcal{T}}(z) = -\log z$, and the variables are freely independent, fits (1.8): for $b > 0$ take $A = a$, $B = b$, and $c = b\log b$, with any extra location added to c. The direction of strict perfect skewness is therefore $-\operatorname{sgn}(b)$ when $b \neq 0$. This covers arbitrary Cauchy perturbations of the exceptional law.

6.2 Prior work and the bounded literature audit

Biane's density coordinates and unimodality theorem are established background [1]. Hasebe–Kuznetsov [2] provide the transform normalization used here. Corpus [122] [7] already uses the angular sine bound to study the mean, median and mode, so those ingredients are explicitly credited above. The exact target of the present proof is the general question in [3, Remark 13], not the mean–median–mode question in its following remark.

The literature audit was carried out on 8 November 2026. It included the primary sources [1, 2, 3], the May 2018 version [4], the related 2026 preprint [6], and the 130 TEX manuscripts in the supplied corpus at the commit recorded in [7]. In the inspected material we did not locate an earlier general perfect-skewness theorem or the same combined proof. Wang's 2026 preprint concerns classical divisibility and density-shape questions; its main results and relevant sections were inspected for overlap, without using its theorems as premises of our proof.

This is a bounded audit, not a proof that no antecedent exists. Accordingly, mathematical completeness of the argument presented here, independent external verification, and historical priority remain distinct claims. The manuscript is a complete proof draft for external review; it does not report acceptance or verification by a journal or by the original question's authors.

6.3 The classical problem is a separate target

The classical stable analogue is associated with Hall's paper [5] in the discussion of [3, Remark 13]. The present proof uses the inverse *free* Cauchy transform (1.2). It provides no proof of the corresponding classical stable-density statement. In particular, an affirmative answer to the free question must not be described as a resolution of Hall's classical problem.

A Checks against the requested success conditions

The following audit records what is established by the manuscript and what the bibliographic search cannot certify.

Gate	Requirement	Evidence and status
G1	A pre-existing identified problem	The target is the explicit perfect-skewness question in Hasebe–Simon–Wang, Remark 13 (2020), already in the 2018 preprint as Remark 18. The reference and original density comparison are given in Section 1.
G2	The original statement is unchanged	Theorem 1.1 proves the same comparison for all admissible parameters and all $t > 0$, and specifies its sign. Lemma 2.3 proves the implication from the auxiliary midpoints to that original statement.
G3	The problem was actually unresolved	The original source poses the general case as open. No later general solution was located in the inspected primary material. The bounded audit does not certify the absence of every intervening result.
G4	An unconditional complete proof	Sections 2–5 close the analytic argument for all stated parameters and levels, including the support edges. No conjecture, numerical experiment, or finite-height exclusion is a premise. External mathematical review remains pending.
G5	A substantive step beyond reparametrization	The strict equal-level curvature gap (2.4), the squared-slope integral (2.9), and the midpoint identities (4.8) and (5.7) supply the global sign. The earlier coordinates and sine bound are credited.
G6	No earlier identical solution	No duplicate theorem with the same general scope or essentially identical argument was found in the audited sources. Historical priority is not certified by search absence.

Quantifier audit. The proof includes $0 < \alpha < 1$ and $1 < \alpha < 2$, both extremal admissible skewness parameters, the symmetric case, the nondegenerate strictly stable index-one cases, $\alpha = 2$, and every logarithmic index-one law after reflection. All density heights $0 < h < h_*$ and all distances $t > 0$ are covered. Point masses are excluded because they have no Lebesgue density.

Reproducibility. This single TEX file contains the complete proof and bibliography. It requires no external data, figure, bibliography file, computer algebra output, or numerical certificate. It compiles with two runs of `pdflatex`.

References

[1] H. Bercovici and V. Pata, with an appendix by P. Biane, *Stable laws and domains of attraction in free probability theory*, Annals of Mathematics (2) **149** (1999), 1023–1060. doi:10.2307/121080. arXiv:math/9905206. The density parametrizations used here are in Appendix A.1.

[2] T. Hasebe and A. Kuznetsov, *On free stable distributions*, Electronic Communications in Probability **19** (2014), paper no. 56, 1–12. doi:10.1214/ECP.v19-3443.

[3] T. Hasebe, T. Simon and M. Wang, *Some properties of the free stable distributions*, Annales de l'Institut Henri Poincaré, Probabilités et Statistiques **56** (2020), no. 1, 296–325. doi:10.1214/19-AIHP962. Author-hosted published PDF. The target is Remark 13, p. 321; the index-one classification is discussed on p. 298.

[4] T. Hasebe, T. Simon and M. Wang, *Some properties of the free stable distributions*, arXiv:1805.01133v2, 7 May 2018; first submitted 3 May 2018. The corresponding question is Remark 18, p. 39, in this version.

[5] P. Hall, *On unimodality and rates of convergence for stable laws*, Journal of the London Mathematical Society (2) **30** (1984), no. 2, 371–384. doi:10.1112/jlms/s2-30.2.371.

[6] M. Wang, *Classical infinite divisibility, self-decomposability and bell-shape of free stable laws*, preprint, 9 October 2026, Hexagon 2610.00169v1, 77 pp. Primary PDF. Inspected for overlap; no result from this preprint is assumed in the proofs above.

[7] *Mean, median and mode of stable laws, and a sharp gamma-mixture criterion*, research manuscript prepared with ChatGPT for Keio, 3 November 2026, item [122] in *Verifiable Mathematical Reasoning Corpus*, repository maintained by QEEDDD. Corpus snapshot: commit c30d401. File: `Mean_Median_Mode_and_Gamma_Mixtures_of_Stable_Laws.tex` (with item prefix [122]).